

49-Day War Plan: Nano-Rare Gene Therapy Framework

Objective: Build a publishable, generalizable computational framework that matches untreated nano-rare monogenic diseases to gene therapy development strategies via multi-parameter structural homology and clinical surrogate matching.

Final Deliverable: An 8000-word manuscript with validated pipeline, retrospective controls, prospective case studies, and 70 references.

Timeline: 49 days (7 weeks), 10-14 hours/day, no weekends off.

Working Directory: /Users/suzie/nano-rare-gt/ (primary repo)

This Plan Location: /Users/suzie/Desktop/GitHub/nano-rare-gt/

1. THE ALGORITHMIC PIPELINE

INPUT: Orphanet ID / OMIM ID / Gene Symbol of untreated nano-rare monogenic disease

OUTPUT: Ranked GT development strategies with feasibility scores, risk flags, auto-generated protocols

Stage 1 — Disease Characterization

- Orphanet API: prevalence, inheritance, phenotypes, severity
- OMIM cross-ref: gene mapping, allelic disorders
- ClinicalTrials.gov: active GT trial count (exclusion gate: >0 active trials deprioritized unless no approved therapy)
- Inheritance flag: homozygous LOF, dominant negative, GOF (determines knockdown vs cDNA vs editing strategy)

Stage 2 — Gene & Protein Annotation

- UniProt API: sequence, domains, GO terms, MW, subcellular location
- AlphaFold DB: structure availability, pLDDT confidence (<50 = low-confidence flag)
- Secreted vs intracellular classifier (secreted = higher cross-correction potential = easier target)

Stage 3 — Structural Homology & Surrogate Matching

- BLASTp / Foldseek vs all proteins with approved/phase-3 gene therapies
- Scores: global sequence identity, domain architecture similarity (Pfam/InterPro overlap), structure alignment RMSD (if both have AF models)
- Surrogate selection: approved GT with highest structural homology + shared tissue tropism

Stage 4 — Vector Compatibility

- CDS length vs AAV packaging limit (4.7 kb hard gate)
- If >4.7 kb: flag oversized, suggest dual-AAV, lentivirus, or mini-gene design
- Serotype matching: target tissue to optimal serotype (CNS -> AAV9/PHP.eB, retina -> AAV2/AAV8, liver -> AAV8/AAV9, muscle -> AAVrh74/AAV9)
- Patent/freedom-to-operate flag on serotypes

Stage 5 — Expression & Regulatory Matching

- GTEx: target tissue TPM (>1.0 on-target, <0.1 off-target avoidance)

- Promoter selection: ubiquitous (CMV/CBA) vs tissue-specific (RHO retina, MHCK7 muscle, TBG liver, GfaABC1D astrocytes)
- TiProD or literature for regulatory element precedents

Stage 6 — Immunogenicity Risk

- IEDB-based MHC-I epitope prediction for transgene vs common HLA alleles (*A02:01*, *A01:01*, *B07:02*, *DRB104:01*)
- Flag high epitope density (>20 predicted binders)
- Compare to human ortholog similarity (self-like = lower risk)

Stage 7 — Therapeutic Window & Codon Optimization

- Codon Adaptation Index (CAI) for target organism
- Overexpression toxicity flag (check OMIM/ClinVar for WT overexpression toxicity)
- Therapeutic index: rescue expression needed vs toxicity threshold

Stage 8 — Multi-Parameter Scoring Engine

- 11 dimensions, each 0-1 normalized
- Weights: structural homology (0.20), size compatibility (0.15, hard gate), tissue tropism (0.15), platform depth (0.15), immunogenicity (0.10), promoter match (0.10), therapeutic window (0.10), codon opt (0.05)
- Confidence tiers: high (all primary data), medium (1-2 gaps), low (major gaps)
- Auto-reject if size gate fails AND no dual-AAV/lenti workaround selected

Stage 9 — Protocol Synthesis

- Jinja2 template: cargo cDNA, vector serotype, promoter, RoA, dosing, risks, mitigations
- Output: JSONL for downstream + Markdown report for human review

2. THE 49-DAY BATTLE PLAN

WEEK 1: FOUNDATION & DATA LAYER (Days 1-7)

Day 1 — Environment Lock - Purge R artifacts from primary repo (.Rhistory, .Rproj) - Finalize VS Code workspace: .venv, black, ruff, mypy - Run full test suite; fix any breakage - Write ADR-001: “Why SQLite for v1, PostgreSQL for v2” - Set up requests-cache with 7-day TTL for all external APIs - Git commit: “w1d1: env locked, caching configured”

Day 2 — Schema Hardening - Review existing models.py; add missing fields if needed - Finalize SQLite schema in db.py; ensure clean serialization - Add Alembic migration baseline - Test: roundtrip Disease -> DB -> Pydantic -> dict - Git commit: “w1d2: schema hardened”

Day 3 — Orphanet Pipeline - Build orphanet.py client: query by ID, parse prevalence/severity/inheritance - Graceful API failure handling (cached fallback + error logging) - Map phenotypes to HPO terms if possible - Test with mocked response - Git commit: “w1d3: orphanet client”

Day 4 — OMIM & ClinicalTrials.gov - Build omim.py: gene mapping, allelic disorders - Build trials.py: count active GT trials by disease/gene query - Implement exclusion gate: active trial count >

0 -> flag “existing development” - Integration test: Orphanet ID -> OMIM gene -> trial count - Git commit: “w1d4: omim + trials linkage”

Day 5 — UniProt Integration - Build uniprot.py: fetch sequence, domains, GO, MW, subcellular location - Cache raw XML/JSON in data/cache/ - Implement Protein model population - Build secreted/intracellular classifier from go_terms and keywords - Git commit: “w1d5: uniprot pipeline”

Day 6 — AlphaFold DB Linkage - Build alphafold.py: map UniProt ID -> AFDB ID -> structure URL - Download PDB/CIF for validation cases to data/structures/ - Implement pLDDT confidence extraction - Flag avg pLDDT < 50 as low-confidence - Git commit: “w1d6: alphafold linkage”

Day 7 — Week 1 Integration - E2E test: ORPHA:324 -> full disease/gene/protein/structure record in <30 seconds - Profile API call times; optimize slow paths - ruff + mypy: zero warnings - Update README with Week 1 progress - Git commit: “w1d7: disease characterization pipeline complete”

WEEK 2: HOMOLOGY, VECTOR, EXPRESSION (Days 8-14)

Day 8 — Surrogate Database Construction - Manually curate JSON/CSV of approved + phase-3 gene therapies - Fields: program_name, target_gene, vector_serotype, promoter, target_tissue, cds_length_bp, disease_orphanet_id - Target: 20-30 entries minimum (Luxturna, Zolgensma, Hemgenix, Roctavian, Elevidys, etc.) - Store in data/surrogate_db.jsonl - Git commit: “w2d8: surrogate db v1”

Day 9 — Sequence Homology Module - Build homology.py: BLASTp wrapper or EBI NCBIblast API client - BLAST query protein against all surrogate_db sequences - Score: % identity, % similarity, alignment coverage - Unit tests with known homologs - Git commit: “w2d9: sequence homology”

Day 10 — Structural Homology & Domain Matching - Build structure.py: compute domain overlap (Pfam/InterPro intersection/union) - If both have AF models: use Foldseek/TM-align if installable; else use domain architecture score - Weight: sequence identity 0.6, domain similarity 0.4 for structural score - Git commit: “w2d10: domain + structural scoring”

Day 11 — Vector Sizing & Serotype Matching - Build vector.py: gate on cds_length_bp vs cargo_limit_bp (4700 bp hard) - If >4700 bp: flag oversized, suggest dual-AAV/lenti/minigene - Match tissue to serotype from data/serotype_map.json - Score tissue_tropism: exact=1.0, related=0.5, no match=0.0 - Git commit: “w2d11: vector compatibility engine”

Day 12 — Tissue Expression (GTEx) - Build expression.py: GTEx API or local GTEx median TPM parsing - Fetch expression across 50+ tissues for query gene - Map primary target tissue from disease phenotype terms - Score: high target-tissue + low off-target = high score - Git commit: “w2d12: gtex expression matching”

Day 13 — Promoter & Regulatory Matching - Build promoter.py: static knowledge base tissue -> recommended promoters - Sources: literature, TiProD, GTEx co-expression - Score promoter match: exact tissue-specific=1.0, ubiquitous=0.6, mismatched=0.2 - Git commit: “w2d13: promoter selection”

Day 14 — Week 2 Integration - Pipeline test: query gene -> structural homology -> vector match -> expression -> promoter - Benchmark: 5 diseases in <5 minutes - Zero lint warnings - Git commit: “w2d14: matching engine v1”

WEEK 3: IMMUNOGENICITY, SCORING, CLI, REPORTS (Days 15-21)

Day 15 — Immunogenicity Module - Build immuno.py: IEDB API or local MHC-I binding prediction - If installable: mhcflurry or netMHCpan; else use sequence heuristics - Flag high epitope count; compare to human self-similarity - Git commit: “w3d15: immunogenicity risk”

Day 16 — Codon Optimization & Therapeutic Window - Build codon.py: calculate CAI using standard human codon usage table - Build therapeutics.py: check OMIM/ClinVar for WT overexpression toxicity - Score therapeutic window: 1.0 no toxicity, 0.5 uncertain, 0.0 known toxicity - Git commit: “w3d16: codon + toxicity”

Day 17 — Scoring Engine v1 - Build scoring.py: implement ScoreBreakdown population from all modules - Normalization: raw score -> 0-1 scale with disease-aware percentiles - Hard gates and composite weighting (ADR-002) - Unit test: known match scores higher than random pair - Git commit: “w3d17: scoring engine”

Day 18 — Ranking & Confidence - Build ranking.py: sort by composite score, break ties by platform depth - Confidence tiers: high/all primary data, medium/1-2 gaps, low/>2 gaps - Top match selection with justification extraction - Git commit: “w3d18: ranking + confidence”

Day 19 — Report Generation - Build/refine report.py: Jinja2 templates for Markdown output - Sections: Executive Summary, Disease Profile, Surrogate Rationale, Protocol Recommendation, Risk Assessment, Development Roadmap - Every score dimension gets 1-2 sentences justification - Git commit: “w3d19: report templates”

Day 20 — CLI Wiring - Wire all modules into cli.py: nanogt match -disease ORPHA:XXX -output report.md - Add -format json, -top-n - Rich progress bars for long queries - Git commit: “w3d20: CLI wired”

Day 21 — End-to-End Pole Test - Run full pipeline on 5 diverse diseases (CNS, retina, liver, muscle, metabolic) - Verify reports are coherent and scientifically defensible - Fix all bugs. Zero mypy warnings. - Git tag: v0.2-alpha - Git commit: “w3d21: e2e validated”

WEEK 4: VALIDATION & CASE STUDIES (Days 22-28)

Day 22 — Retrospective Positive Control #1: Luxturna (RPE65) - Input: RPE65-related retinal dystrophy - Expected: high match to itself, AAV2, subretinal, CMV/CBA - Validate algorithm. Fix discrepancies. - Git commit: “w4d22: luxturna validation”

Day 23 — Retrospective Positive Control #2: Zolgensma (SMN1) - Input: Spinal muscular atrophy - Expected: AAV9, intrathecal/IV, scAAV-CB - Test gene duplication edge case handling - Git commit: “w4d23: zolgensma validation”

Day 24 — Retrospective Negative Control - Research one discontinued GT program (e.g., early CNS failures, certain hemophilia AAV5 programs with immunogenicity issues) - Input target gene/disease. Algorithm should flag immunogenicity risk or tissue mismatch matching known failure mode. - Git commit: “w4d24: negative control validation”

Day 25 — Prospective Case Study #1: CNS Ultra-Rare - Select untreated ultra-rare CNS monogenic disease with homozygous LOF - Run full pipeline. Generate protocol. Manually verify against literature. - Write 800-word case study narrative. - Git commit: “w4d25: prospective cns case”

Day 26 — Prospective Case Study #2: Metabolic/Hepatic - Select untreated metabolic liver disease - Run pipeline. Focus on AAV8/9 liver targeting, TBG promoter. - Verify CDS fit and GTEx

liver expression. - Write 800-word case study narrative. - Git commit: “w4d26: prospective hepatic case”

Day 27 — Sensitivity Analysis - Systematically vary scoring weights +/- 20% and re-run retrospective cases - Measure rank stability. Report which dimensions are most influential. - Generate weight sensitivity heatmap data for paper figures. - Git commit: “w4d27: sensitivity analysis”

Day 28 — Case Study Documentation - Compile all validation into docs/validation.md - Include expected vs observed, algorithm fixes, limitations - Generate first draft schematics - Git commit: “w4d28: validation complete”

WEEK 5: PAPER WRITING — CORE SECTIONS (Days 29-35)

Day 29 — Introduction & Background (target 1200 words) - Nano-rare disease definition and prevalence statistics - Gap: thousands of monogenic diseases, <50 approved GTs - Solution: computational framework leveraging structural homology + clinical surrogates - Prior art and hypothesis - File: paper/introduction.md

Day 30 — Methods — System Overview (target 600 words) - Architecture diagram description - Data sources and APIs - Software stack: Python, Pydantic, SQLite, Typer - Ethical/data statement: all public data, no patient data - File: paper/methods_system.md

Day 31 — Methods — Disease & Gene Module (target 400 words) - Orphanet + OMIM + ClinicalTrials.gov integration - Disease severity scoring, trial-status exclusion - File: paper/methods_disease.md

Day 32 — Methods — Homology & Vector (target 500 words) - BLASTp parameters, domain similarity metric, surrogate DB construction - AAV sizing gate, serotype selection, packaging limit handling - File: paper/methods_homology.md

Day 33 — Methods — Expression & Immunogenicity (target 500 words) - GTEx thresholds, promoter selection rules - MHC-I epitope prediction approach, self-similarity correction - Codon optimization (CAI), therapeutic window assessment - File: paper/methods_expression.md

Day 34 — Methods — Scoring & Output (target 400 words) - Weighted composite formula, normalization, confidence tiers - Protocol synthesis via template generation - File: paper/methods_scoring.md

Day 35 — Assemble Methods - Combine all methods into paper/methods_full.md - Cross-references, consistent notation - Internal word count check - Git commit: “w5d35: methods draft complete”

WEEK 6: RESULTS, DISCUSSION, FIGURES (Days 36-42)

Day 36 — Results — Pipeline Performance (target 800 words) - Database coverage statistics - Query speed benchmarks (median, 95th percentile) - Data completeness per dimension - File: paper/results_performance.md

Day 37 — Results — Retrospective Validation (target 700 words) - Luxturna, Zolgensma results with scores - Negative control discrimination - Accuracy metrics: top-k retrieval, rank correlation - File: paper/results_validation.md

Day 38 — Results — Prospective Case Studies (target 800 words) - CNS and hepatic cases with full protocols - Score breakdown tables - File: paper/results_cases.md

Day 39 — Figures - Figure 1: Pipeline schematic (SVG/PNG, high-res) - Figure 2: Scoring radar chart for top case study - Figure 3: Sensitivity analysis heatmap (weight perturbation vs rank) - Figure 4: Composite score distribution across tested diseases - Figure 5: Prospective case study comparison table + schematic - All at 300 DPI, journal-ready - Git commit: “w6d39: figures finalized”

Day 40 — Discussion — Strengths & Limitations (target 1000 words) - Generalizability across monogenic disease classes - Limitations: AlphaFold confidence, incomplete surrogate DB, immunogenicity prediction accuracy, lack of patient-specific HLA data - Comparison to manual GT development timelines - File: paper/discussion_strengths.md

Day 41 — Discussion — Regulatory, CMC, IP & Future (target 1000 words) - Regulatory pathway acceleration via surrogate precedent - CMC: platform manufacturing vs novel cargo - IP: freedom-to-operate for serotypes, promoters, transgenes - Future work: CRISPR expansion, non-viral delivery, ML refinement - File: paper/discussion_future.md

Day 42 — First Full Draft Assembly - Combine all sections into paper/manuscript_v1.md - Add Abstract (250 words), Keywords, Title - Target total: 7500-8000 words excluding references - Read aloud, fix logic gaps, check transitions - Git commit: “w6d42: manuscript v1 complete”

WEEK 7: REFINEMENT, POLISH, SUBMISSION PREP (Days 43-49)

Day 43 — Citation Audit - Ensure every claim has a citation - Add 60-80 references: OMIM/Orphanet methods, GT reviews, AAV engineering, AlphaFold, GTEx, IEDB, prior approvals - Decide target journal: Molecular Therapy (Methods & Clinical Development) for methods focus, or Nature Communications for broader scope - Consistent formatting (Vancouver for MolTher; APA for Nature) - File: paper/references.bib or formatted text - Git commit: “w7d43: citations complete”

Day 44 — Code Reproducibility Audit - Re-run all analyses from fresh clone / fresh venv - Document exact dependency versions (pip freeze > requirements.txt) - Ensure README has installation + usage instructions - Create docs/reproducibility.md with expected runtimes and outputs - Git commit: “w7d44: reproducibility verified”

Day 45 — Language Edit - Reduce jargon where possible - Replace passive voice where active is stronger - Frame: “generalizable framework applicable across the full spectrum of monogenic disease” (scientifically defensible) - Check word count exactly on target (7500-8500) - Git commit: “w7d45: language edit”

Day 46 — Journal Formatting & Supplementary - Format per chosen journal guidelines (font, headings, refs) - Supplementary tables: full scoring tables, surrogate DB as Supp Table 1, weight justification as Supp Note 1 - Git commit: “w7d46: supplementary complete”

Day 47 — Peer Pre-Review (Self-Critique) - Read as Reviewer #2. Anticipate: - “Is the surrogate database large enough?” - “How do you handle genes >4.7kb?” - “What about gene editing approaches?” - Write responses in paper/anticipated_reviews.md - Adjust manuscript to address top 3 criticisms preemptively - Git commit: “w7d47: pre-review pass”

Day 48 — Final Figures & Accessibility - All figures meet journal resolution requirements - Alt-text for accessibility - Generate final PDF from Markdown (pandoc or preferred tool) - Generate code archive zip for supplemental software - Git commit: “w7d48: final outputs”

Day 49 — Final Read-Through & Submission Prep - Read entire paper start-to-finish without editing - Write cover letter: novelty, significance, audience - Check supplementary package completeness - Submit to target journal OR post to bioRxiv - Final git tag: v1.0-submission - Git commit: “w7d49: SUBMISSION READY”

3. DAILY SCHEDULE (Mad Scientist Mode)

Time	Activity
0600	Wake, coffee, 30 min paper reading (Nature / Molecular Therapy / GT news)
0630	3-hour deep work block (no Slack, no email, no phone)
0930	Break, walk, snack
1000	3-hour deep work block
1300	Lunch + 30 min break (walk outside, no screens)
1400	2-hour coding/writing block
1600	Coffee + 30 min break
1630	2-hour integration/testing/writing block
1830	Dinner break
1930	2-hour lighter work: documentation, tests, reading, citations
2130	Git commit, write tomorrow's 3 specific objectives
2200	Stop. Sleep is non-negotiable for cognitive performance.

Total: ~12 hours focused work daily. Sustainable for 7 weeks if sleep and exercise are protected.

4. CONTINGENCY PROTOCOLS

Problem	Response
API outage (Orphanet/UniProt/CT.gov down)	Use local cached data (requests-cache 7-day TTL). If >2 days, build static JSON from cache and document limitation in Methods.
No AlphaFold structure	Fallback to domain architecture-only scoring. Document lower confidence tier.
Gene > 4.7kb (AAV limit)	Do not discard. Branch: suggest dual AAV, AAV-ITR minigene, or lentivirus. Document as “oversized cargo pathway.”
Immunogenicity prediction too slow	Simplify to epitope count threshold rather than full MHC binding for v1. Document in limitations.
Writer's block	Switch to figure-making or test-writing for 1 day. Return to prose after.
Mypy/ruff refuses to pass	Spend max 2 hours; if stuck, document the type ignore in an ADR and move on.
Surrogate DB <20 entries by Day 9	Lower threshold to 15 + document as v1 limitation. Expand to 30+ in Week 4 as background task.

5. SUCCESS CRITERIA (Day 49)

- ☐ Repository: 2000+ lines of tested Python, 80%+ coverage
- ☐ Paper: 8000 words, 5 figures, 2 tables, 70 references, formatted for target journal

- ☐ Validation: 2 retrospective positive controls, 1 negative control, 2 prospective cases
 - ☐ Deliverable: Complete GT protocol for untreated ultra-rare disease in <2 minutes via `nanogt match --disease ORPHA:XXX`
 - ☐ Reproducibility: fresh install + run produces identical results in <30 minutes setup
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6. TARGET JOURNAL OPTIONS

1. **Molecular Therapy — Methods & Clinical Development** (methods-focused, rapid review)
2. **Nature Communications** (broad reach, high impact)
3. **Human Gene Therapy** (specialist audience)
4. **bioRxiv** (immediate preprint if journal submission feels premature)

Decide on **Day 43** after reading recent issues of each.

7. FILES TO CREATE NOW

After reading this plan, create these files in your repo TODAY:

```
cd /Users/suzie/nano-rare-gt
mkdir -p paper docs/figures data/structures data/cache tests/fixtures
```

This plan was generated on 17 May 2026. The clock starts when you say go.